

B9 Chop

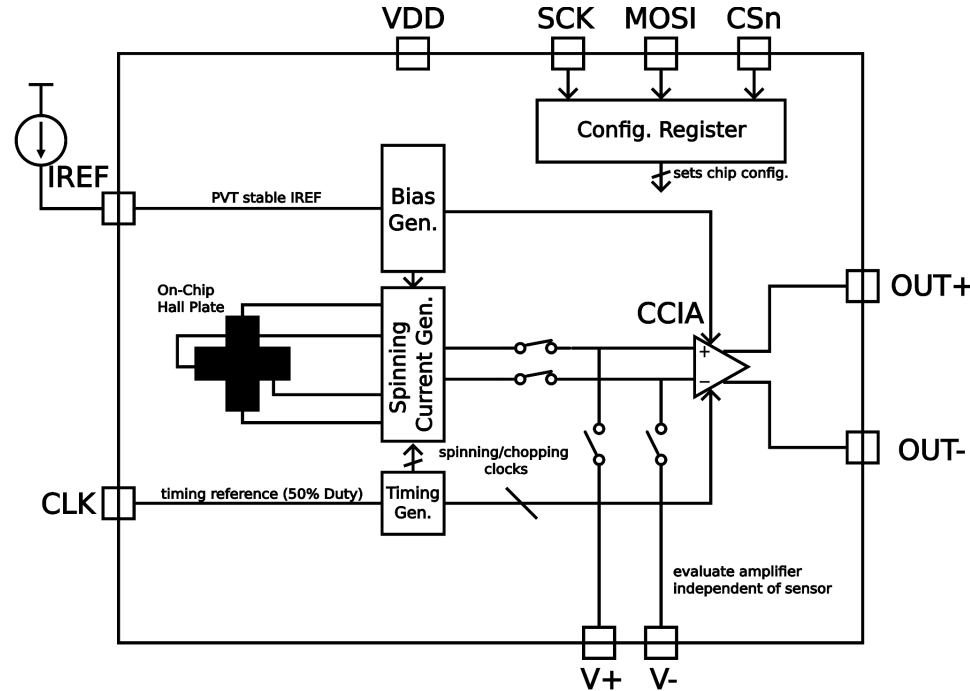
CCIA for Linear Hall Effect Sensor

Team members (not read in presentation)

Discord name	Affiliation	Experience	Role
alang	University of Florida	Master's Student	Team lead

Project information

- Goal
 - Enable monolithic magnetic sensing applications in GF180 technology
 - Build and evaluate sensor & front-end for future projects (e.g., Hall sensor array, digital read-out)
- Design (Pad Frame D - 10 pin, 1/16)
 - Spinning current method [1] used to bias on-chip sensor (modulates hall voltage)
 - CCIA [2] amplifies and demodulates Hall voltage
 - Spinning removes sensor/amplifier offset and 1/f noise
 - Simple SPI configuration register used to control on-chip switches to enable several test configurations (test amplifier with voltage input, test sensor with off-chip amplifier, vary sensor bias current, etc.)



Project information

- Application

- Many applications for monolithic magnetic sensors
 - Contact-less Current Sensing [3]
 - Position/Velocity Sensing [4]
 - Biosensors [5]
- Chipathon implementation will target lower bandwidth sensor (~ 10 kHz) to reduce complexity

- References

- **[1]** V. Mosser, N. Matringe and Y. Haddab, "A Spinning Current Circuit for Hall Measurements Down to the Nanotesla Range," in IEEE Transactions on Instrumentation and Measurement, vol. 66, no. 4, pp. 637-650, April 2017, doi: 10.1109/TIM.2017.2649858.
- **[2]** Q. Fan, F. Sebastiano, J. H. Huijsing and K. A. A. Makinwa, "A $1.8\ \mu\text{W}$ 60 nV/Hz Capacitively-Coupled Chopper Instrumentation Amplifier in 65 nm CMOS for Wireless Sensor Nodes," in IEEE Journal of Solid-State Circuits, vol. 46, no. 7, pp. 1534-1543, July 2011, doi: 10.1109/JSSC.2011.2143610.
- **[3]** M. Crescentini, M. Marchesi, A. Romani, M. Tartagni and P. A. Traverso, "A Broadband, On-Chip Sensor Based on Hall Effect for Current Measurements in Smart Power Circuits," in IEEE Transactions on Instrumentation and Measurement, vol. 67, no. 6, pp. 1470-1485, June 2018, doi: 10.1109/TIM.2018.2795248.
- **[4]** B. Wang, X. Lai, Z. Niu and L. Li, "A Fully Integrated Chopped Hall Sensor for Gear Rotation Measurements," in IEEE Transactions on Instrumentation and Measurement, vol. 74, pp. 1-11, 2025, Art no. 2001311, doi: 10.1109/TIM.2024.3522378.
- **[5]** S. Gambini, K. Skucha, P. P. Liu, J. Kim and R. Krigel, "A 10 kPixel CMOS Hall Sensor Array With Baseline Suppression and Parallel Readout for Immunoassays," in IEEE Journal of Solid-State Circuits, vol. 48, no. 1, pp. 302-317, Jan. 2013, doi: 10.1109/JSSC.2012.2224531.